Product & Engineering Execution Report

Case Study: Fullstack Product Engineer (Frontend Revamp Depth) — AI Interview Platform

Platform: AI Interview Platform (Rails 7 + React 18 TypeScript)

Repository: github.com/rakamindev/ai-interview-platform

Feature Branch: `feature/frontend-product-revamp`

Seeded Fault Branch: `scratch/seeded-fault-test`

Evaluation Ratio: 50% Product Impact & Craft • 50% Engineering Execution

Status: Ready for Production & Client Release

1. Executive Summary & Pull Request Link

This report documents the end-to-end audit, problem definition, architectural revamp, and automated test suite for the **AI Interview Platform**. Treated as software about to ship to enterprise clients, this initiative directly resolves critical multi-tenant data leaks (UU PDP compliance), eliminates unfair unassessed skill rating distortions (preventing false L1 ratings), and bridges the frontend/backend data contract seam on the Fit/Gap comparison matrix.

> **GitHub Pull Request:** https://github.com/rakamindev/ai-interview-platform/pull/212
> *All changes are organized in atomic, descriptive commits across both services with 100% test pass rate and clean build.*

2. Step 2: Deep Context & Domain Immersion

A. Indonesian Hiring Context & Assessor Pain Points

Technical hiring in Indonesia suffers from acute engineering resource exhaustion. Senior software engineers frequently spend 15–20 hours each week conducting repetitive initial screening interviews. Assessors and hiring managers require a transparent, reliable competency signal—not static theoretical recitation—to benchmark talent quickly against vacancy requirements.

B. High-Stakes Candidate Empathy

Candidates do not select the AI tool and cannot opt out. An arbitrary evaluation failure or a confusing user interface alters a real person's career trajectory. When unassessed skills default to failing grades (L1) due to early interview termination, the platform actively harms the candidate. Designing with candidate empathy requires explicit handling of incomplete sessions, transparent override indicators, and graceful degradation during network drops.

C. Regulatory Compliance (UU No. 27 Tahun 2022 - UU PDP)

Transcripts, voice recordings, and evaluator ratings constitute sensitive personal data under Indonesia's Personal Data Protection Law (UU PDP). Strict multi-tenant isolation is mandatory. An assessor from Organization A must never be able to inspect, query, or override candidate evaluation records belonging to Organization B.

3. Step 3: Defining Problem & Gap Analysis (P0 to P3)

Severity	Finding	One-Line Impact Statement	Classification
P0	Cross-Tenant Portfolio Access	Assessor from Tenant A can view/override evaluations of Tenant B due to unscoped `Portfolio.find(params[:id])`, causing severe UU PDP data breach.	Defective Implementation
P1	Unassessed Skills Clamped to L1	Unassessed skills (probe_count: 0) execute `.to_i.clamp(1, 5)`, penalizing candidates with a false failing L1 score on topics never explored.	Defective Implementation
P1	Fit/Gap Contract Key Mismatch	Backend serializes `expected_level` while frontend reads `required_level`, rendering the required column completely blank.	Defective Implementation
P1	Missing Assessor Override Flag	Backend never sent `is_override` flag, preventing frontend from indicating whether a score was AI-generated or human-adjusted.	Missing Specification
P2	Zero Automated Test Harness	No test runner existed in `web/` and no integration specs existed in `api/`, risking silent regressions upon deployment.	Missing Specification
P2	Truncated Narrative Presentation	`FitGapReportPage` used `culture_narrative || overall_narrative`, completely hiding the overall recommendation if culture was present.	Defective Implementation

4. Step 4: Revamp Strategy, Acceptance Criteria & Trade-offs

A. Technical Solution Options & Trade-off Evaluation

Dimension	Option A: Frontend-Only Quick Hack	Option B: Full-Stack Resilient Contract & Craft (Chosen)
Product Impact vs Cost	Low cost, but ignores multi-tenant data leaks and background job DB validation crashes.	High impact. Guarantees UU PDP tenant isolation, fair candidate scoring, and rich UX with zero data corruption.
Long-term Maintainability	Fragile. Disconnected schemas require constant client-side patching.	Clean, decoupled interfaces. Dual-field tolerance ensures forward & backward API compatibility.
Failure Modes	Unassessed skills crash Sidekiq workers with enum validation errors.	Graceful degradation: nil/unassessed skills render neutral "Not assessed" badges without throwing errors.

B. Self-Defined Acceptance Criteria (Edge Cases Handled)

- **Unassessed Skills:** If a configured skill has 0 probes, it displays a neutral ⚪ `Not assessed` badge in Fit/Gap and `– / Unassessed` in Portfolio cards. It must never force L1.
- **Contract Resilience:** `ComparisonTable` resolves levels via `c.expected_level ?? c.required_level`, gracefully defaulting to – if omitted.
- **Human Overrides:** Overridden skills show a distinct ✏️ `Override (AI: LX)` tag, preserving auditability between AI and human judgement.
- **Executive KPI Header & Responsive Layout:** Wide `max-w-5xl` layout with Role Benchmark, Match Rate % progress bar, Dynamic Fit Status badge, and Level Badge color pills.
- **Dual Narrative Display:** Culture & Competency Fit and Overall Hiring Assessment render in separate dedicated cards with AI objective synthesis tags.
- **Direct Assessor Navigation:** 1-click direct `Transcript` access from candidate session list for frictionless review.

5. Step 5: Monozukuri Implementation & Testing Proofs

A. Automated Test Suite Execution (Vitest)

We built an automated testing harness using **Vitest** and **React Testing Library** in `web/`. 9 tests across 2 test suites cover all critical components:

```
$ npm test
 RUN  v5.0.5 C:/Users/kirei/ai-interview-platform/web

 ✓ src/components/fitgap/__tests__/ComparisonTable.test.tsx (5 tests)
 ✓ src/components/portfolio/__tests__/SkillPortfolioCard.test.tsx (4 tests)

 Test Files  2 passed (2)
      Tests  9 passed (9)
   Duration  1.94s
```

B. Seeded Fault Test Proof (Preserved on `scratch/seeded-fault-test`)

To prove our test harness catches real regressions (and is not a set of trivial happy-path assertions):

1. Checked out branch `scratch/seeded-fault-test`.
2. Intentionally broke the level fallback in `ComparisonTable.tsx` by ignoring `expected_level`.
3. Ran `npm test`, which immediately caught the fault with a descriptive assertion error:

```
FAIL src/components/fitgap/__tests__/ComparisonTable.test.tsx
  ✕ renders comparisons with expected_level (backend contract) and required_level fallbacks
  Error: expect(element).toHaveTextContent()
  Expected element to have text content: "L3"
  Received: "React / Frontend Architecture—L5⭐Exceeds +2"
```

4. Committed the fault (commit `c948d73`) and reverted it via `git revert` (commit `c8dae98`), restoring the green test suite.

C. AI Verification Moment

The Incident: During initial portfolio logic generation, an AI assistant suggested applying `skill_data['level'].to_i.clamp(1, 5)` to ensure safe integers. However, when an interview ends early and a skill is unassessed, `nil.to_i` returns `0`, and `0.clamp(1, 5)` outputs `1`. This would have stamped an unassessed candidate with an automatic Level 1 (failing grade).
The Monozukuri Resolution: We rejected the clamp and engineered explicit unassessed states: `LevelBadge` displays neutral `– / Unassessed`, and `ComparisonTable` treats missing ratings as `not_assessed`, protecting candidate fairness.

6. Step 6: Video Walkthrough Guide (3–5 Minutes)

Video Demonstration Link: https://youtu.be/3N7zA5cx7LQ?si=IJAAG5ZG76GuEvj7
Walkthrough Structure:

- **0:00–0:45 | Problem Context:** Empathy for candidate stakes and explanation of the broken data seam.
- **0:45–2:00 | Revamped UI Flows:** Showcase the new Fit/Gap matrix, override indicators, and dual narrative layout.
- **2:00–3:15 | Testing Harness:** Run `npm test` and explain the Seeded Fault test on `scratch/seeded-fault-test`.
- **3:15–4:00 | Engineering Craft:** Summary of UU PDP tenant isolation and enterprise-readiness.